

Face Attendance System

Complete Project Roadmap — Simple Language, No Code

Real-time live monitoring · Software built in-house · Existing cameras used first

What we are building. A system where a person walks past a camera, the computer recognises their face, and it records "this person came IN / went OUT at this time" into a database — shown live on a website. No card, no fingerprint, no touching, no stopping to pose.

The one idea behind everything. The camera does not compare photos. It turns every face into **512 numbers**. The same person always makes almost the same numbers; a different person makes very different numbers. Recognising someone = making the numbers for the new face and finding whose saved numbers are the closest. Everything in this roadmap is built around that one idea.

How It Works — The Whole System in 3 Parts

CAMERA + SOFTWARE → DATABASE → WEBSITE
recognises face stores it shows who is in/out

When a person walks in, five things happen in under a second:

Step	What happens
1	The camera sees the face
2	The software turns the face into 512 numbers
3	It compares those numbers to the saved list and finds the person (e.g. Emp 1042)
4	It saves one line: <i>Emp 1042, IN, 08:52</i>
5	The website shows: <i>1042 is now IN</i>

That is the entire system. Everything below is simply how we build these three parts, step by step, in the right order.

Two Things the System Does

	1. ENROL (done once per person)	2. RECOGNISE (every day)
When	Once, when a worker joins	Every time they walk past
Who	HR, in the office	The worker, at the gate
What happens	5 photos taken → turned into 512 numbers → saved against their ID	Face turned into numbers → compared to saved list → punch recorded
Worker effort	Stand and look, 2 minutes	Nothing — just walk past

Enrolment builds the "saved list." Recognition checks against it. You cannot recognise anyone who was never enrolled — so every new joiner must be enrolled on their first day.

The Roadmap — 8 Stages in Order

1 Prove it works (no cost · no camera)

Install the free face-recognition software on any computer. Test it on 5 photos of one office colleague, then a 6th photo. Does it name them correctly? Does it reject a stranger? The moment it does, the hardest technical risk is over.

Done when: The computer correctly names the enrolled person and rejects an unknown one.

2 Build the storage (database) (no cost · no camera)

Create the place where everything is saved. Four simple lists: (1) who the employees are, (2) their face numbers, (3) every in/out event, (4) the daily attendance report built from those events.

Done when: Four empty tables ready to receive data.

3 Build enrolment (no cost · no camera)

Make a simple screen where HR types an ID, captures 5 photos, and saves the person's face numbers. Running this for every worker fills the 'saved list' the camera will check against.

Done when: HR can add a person and their face numbers are stored.

4 Connect the camera (uses existing CCTV)

Point the software at one of the company's existing cameras. A person walks past → the software recognises them → it saves the in/out event automatically. This is the 'live monitoring' part working for the first time.

Done when: Real people walking past are recognised and events appear in the database.

5 Add the attendance rules (no camera · needs HR)

Teach the software the rules: shift timings, how many minutes late is allowed, what counts as a half day, when overtime starts, how to handle night shifts and missing punches. Every rule comes from HR.

Done when: The system correctly shows late, absent, half-day, hours, and overtime.

6 Build the website (dashboard) (no cost · no camera)

The screen management actually looks at: who is currently in, who is out, today's log, and daily/monthly reports — all updating live.

Done when: A live web page showing attendance that anyone authorised can open.

7 Measure accuracy → decide the camera (needs real workers)

Enrol about 50 workers and run the system for 5 working days on the existing camera. Measure how accurate it really is in our light and dust. Use that number — not a guess — to decide which proper camera to buy.

Done when: A real accuracy figure, and a confident camera-buying decision.

8 Run parallel, then go live (final)

Keep the old register running alongside the system for a few weeks. Compare daily. Fix any gaps. Once it matches, stop the register — the system becomes the official record.

Done when: Attendance runs fully on the system. Register retired.

Simple Timeline

Week	What happens	Money
1	Prove it works + start the database	Zero
2	Database done + enrolment screen	Zero
3	Enrol staff + connect existing camera	Zero
4	Attendance rules + start the website	Zero
5–6	Measure accuracy → decide & buy the proper camera	Camera cost
7–10	Run alongside old register, check, then go live	—

The first four weeks cost nothing. We only spend on the camera after we have seen the system actually working on our own people.

Who Does What

Person	Their job
Developer (you)	Build the software, database, and website; run the accuracy test
HR	Give the employee list with line codes + the attendance rules; enrol the workers; keep the list updated when people join or leave
Admin	Fix the camera position and lighting at the gate
PPC / Production	Give planned manpower and daily output (for the manpower reports)
Management	Approve the camera when the accuracy test is done

The 3 Things That Decide Success

Not the AI — that part is free and already very accurate. These three are what actually matter:

Thing	Owner	Why it matters
Employee list with line codes	HR	Turns a plain headcount into "which line has how many people" — the real value for production
Attendance rules	HR	Shift times, late, overtime. Without them the system knows who came but cannot calculate attendance
Camera angle + lighting	Admin	The face must be seen from the front, well lit — not from high above. No software can fix bad light or a bad angle

Important on the camera angle. "Camera at height" is fine, but it must point mostly straight at the face, not steeply down. A camera looking down from directly overhead sees only the top of the head — the face features it needs are hidden. Mount it ahead of the walkway at face level, tilted only slightly down, so people walk toward it.

What the System Uses (Plain List)

All of it is free and open-source. Nothing here needs a licence.

Part	What it is	What it does for us
InsightFace	Free face-recognition software	Finds the face and turns it into the 512 numbers
The face model (ArcFace)	An AI already trained on millions of faces	Makes the 512 numbers reliable — same person close, different far
OpenCV	Free video software	Reads the live camera picture
PostgreSQL	Free database	Stores employees, face numbers, events, attendance
FastAPI	Free backend tool	Connects the software to the website
React	Free website tool	Builds the live dashboard

We do not build the AI. We download it, ready-made. Our real work is the enrolment screen, the attendance rules, the reports, and the website around it.

About the Camera Decision

We deliberately build the software **first**, on existing cameras, and decide the camera **later**. Two reasons:

- Most of the software is the same whichever camera we choose — so no work is wasted.
- By the time we decide, we will have a real accuracy number from our own gate — we buy with proof, not a guess.

For a walk-past system, the camera should be

Mounted ahead of the walkway at roughly face level, tilted only slightly down

Zoomed so the face fills enough of the picture (not the whole gate and road)

Good in low light (night vision / good in shade)

Placed where there is no bright window or sun behind the person

What To Do Next

1. **Today:** install the free software and prove it recognises 5 photos of a colleague (Stage 1).
2. **This week:** build the database and the enrolment screen (Stages 2–3).
3. **Get from IT/Admin:** access to one existing camera's live feed (unblocks Stage 4).

4. **Get from HR:** the employee list with line codes, and the attendance rules — these are the slowest items and block Stages 5 and the manpower reports.
5. **After 50 workers are enrolled:** run the 5-day accuracy test, then decide the camera (Stage 7).

Bottom line. Build the software first on existing cameras — it costs nothing and proves everything. The two real bottlenecks are HR's data and the camera lighting, not the technology. Start Stage 1 today; it needs only a computer and one afternoon.